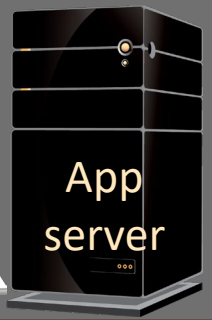
What code goes where?









Application | Application & DB | DB

Various kinds of code

- Application logic
- Business logic – core computation
- Data integrity – error-checking
- Application-DB interface API

Bank example:

- Application logic
  - Website/mobile UI
  - User log-in
- Business logic
  - Creating accounts
  - Computing interest
  - Transactions between accounts

Where does application logic go?



UI, end-user login, … doesn’t belong in DB.

What about the business logic?

Business logic: Application-side

View DB as dumb storage. Data-integrity & API in application.

- Add application servers to handle load.

- Support multiple DBMS (at once or over time)
  - Application is DB-agnostic.
  - DBMS are commodities. Use whichever is currently best.
  - Put logic in the single application, rather than replicating in multiple DBMS.
- Application development tools better (e.g., unit testing).
- Personnel are application-focused.
  - Few good SQL programmers.
  - DB programmers often in a separate DB team.

Business logic: DB-side

DB in charge of data – including data integrity, data computation

- Use functions, triggers extensively
- Application accesses DB via DB-level API
- Add DB servers & replication to handle load

- Application only gets the data it really needs
  - Minimize communication.
  - Expose minimum of DB data & structure. Maximize security.
- DB is application-agnostic.
  - Applications change more frequently than DB architectures.
- Application programmers shouldn't write SQL or pseudo-SQL (ORM).
  - Protect data from bad programmers.
  - Maximize query optimization.

Business logic: Trade-off summary

Application

- **Convenience** – More Java/C++/… programmers than for SQL. Better language, libraries, & tools.
- **Flexibility** – Application controls DB.

DB

- **Encapsulation** – Application programmer only uses API, and doesn't need details, including much SQL.
- **Safety** – Protect DB from application programmer. Only expose desired API.
- **Performance** – Less I/O. Better DBMS optimization.

Business logic: Compromise

Use best features of each:

- DB – CRUD, data integrity, data replication
- Business logic placed as situation needs.
- Application – Everything else

http://www.vertabelo.com/blog/notes-from-the-lab/business-logic-in-the-database-yes-or-no-it-depends